

Exam Date & Time: 26-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - Make up Examination - December -2024

Machine Learning [IT-4032]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and contrast classification and regression tasks in machine learning. Provide examples of each type of task and discuss the specific objectives and evaluation metrics associated with them (5)
 - A)
 - B) Evaluate the effectiveness of ensemble methods such as bagging and boosting in improving the robustness and accuracy of machine learning models. Discuss how these methods combine multiple base learners to overcome individual weaknesses. (5)
- 2) The Expectation-Maximization (EM) algorithm is often used for Gaussian Mixture Models (GMM) in clustering. Outline the mathematical steps of the EM algorithm applied to GMM. How does the EM algorithm determine the parameters of the Gaussian distributions in each cluster, and what are its limitations in handling high-dimensional data? (5)
 - A)
 - B) Explore the concept of kernel trick and its significance in SVM. How does the kernel trick enable SVM to efficiently handle high-dimensional feature spaces without explicitly computing the transformation? Can you provide insights into the mathematical intuition behind the kernel trick. (5)
- 3) Assess the limitations of K-means clustering and propose alternative clustering algorithms that address these limitations. Evaluate the performance of the proposed algorithms in terms of scalability, robustness, and ability to handle non-linear data distributions. (5)
 - A)
 - B) In logistic regression, model coefficients represent the log-odds of the outcome. Describe how to interpret these coefficients in terms of odds ratios. What considerations are necessary when interpreting odds ratios for continuous versus categorical variables, and how do interactions between variables complicate this interpretation? (5)
- 4) Overfitting is a common problem in MLPs trained with back-propagation, especially with small datasets. Analyze how genetic algorithms can be used to select or evolve the architecture and hyperparameters of an MLP to mitigate overfitting. What specific genetic strategies (e.g., cross-validation fitness, regularization) are effective in producing a robust neural network model. (5)
 - A)
 - B) Explain the curse of dimensionality and how it affects the K-NN classifier in high-dimensional datasets. What are the primary challenges caused by high-dimensional

spaces for K-NN, and what techniques (e.g., dimensionality reduction) can be used to mitigate these issues. Discuss the trade-offs involved in applying such techniques.

- 5) Evaluate the impact of data imbalance on the performance of classification algorithms such as logistic regression and support vector machines. Discuss techniques for handling imbalanced datasets and their implications for model fairness and interpretability. (5)

A)

B)

Consider the tuple $X = (\text{Weather} = \text{Rainy}, \text{Temperature} = \text{Mild}, \text{Humidity} = \text{High}, \text{Windy} = \text{True})$. Calculate the conditional probabilities required for predicting the class label 'Outcome' for both 'Win' and 'Loss.'

Weather	Temperature	Humidity	Windy	Outcome
Sunny	Hot	High	False	Win
Sunny	Hot	High	True	Win
Overcast	Hot	High	False	Win
Rainy	Mild	High	False	Win
Rainy	Cool	Normal	False	Lose
Rainy	Cool	Normal	True	Lose
Overcast	Cool	Normal	True	Win
Sunny	Mild	High	False	Win
Sunny	Cool	Normal	False	Win
Rainy	Mild	Normal	False	Lose

(5)

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